山东大学 计算机科学与技术 学院

计算机视觉 课程实验报告

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实验题目：实验 E7：图像匹配 2		

实验过程中遇到和解决的问题：

（记录实验过程中遇到的问题，以及解决过程和实验结果。可以适当配以关键代码辅助说明，但不要大段贴代码。）

1、实验总览

本实验选择对比 SIFT 和 SuperPoint 在同一场景不同视角、同一场景旋转，同一场景缩放下匹配特征点的效果。

2、遇到的问题：

使用的 SuperPoint 项目地址为 https://github.com/rpautrat/SuperPoint

SuperPoint 项目并没有给出 pytorch 版本的使用说明。SuperPoint 原项目使用的 Tenserflow 版本已被弃用。

解决措施：通过查询博客和使用大模型工具，总结得出项目的使用方法。

3、实验原理：

SIFT 是一种经典的特征检测与描述算法。核心目标是提取图像中对尺度缩放，旋转，亮度变化保持稳定的关键点，并生成具有高区分度的特征描述符。SIFT 主要分为尺度空间极值检测、关键点定位、方向分配、描述符生成四部分。

SuperPoint 是一种基于深度学习的特征点检测与描述符生成模型。与传统手工设计特征 SIFT 不同的是，SuperPoint 通过神经网络自动学习图像中的关键点位置及其对应的描述符，显著提升了在复杂场景下的鲁棒性和匹配精度。

结果分析与体会：

对于 SIFT 和 SuperPoint 对比实验，每一组对比实验的输入数据相同。

同时在每一组对比实验中，记录了成功匹配的特征点个数。

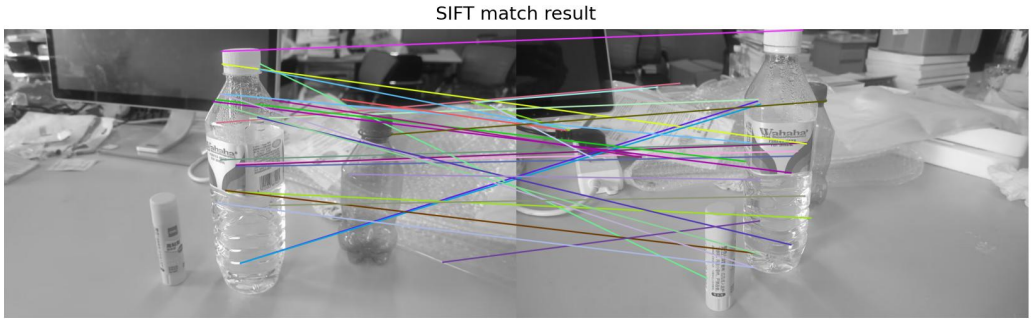
1、对于同一场景，不同角度

原始图片如下两图：





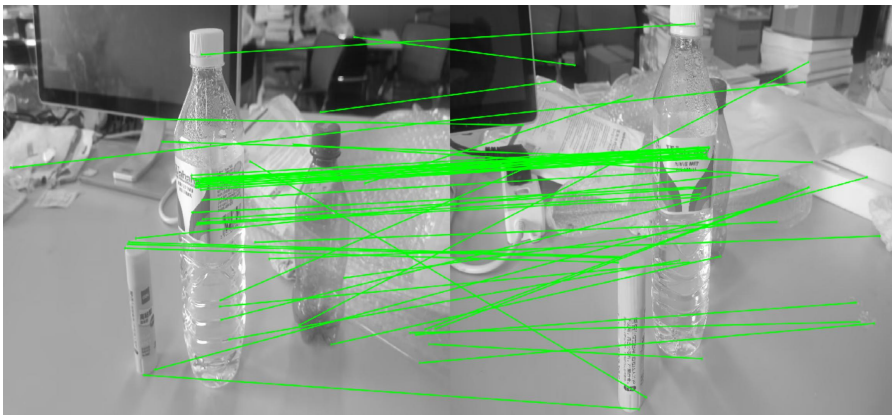
（1）SIFT 结果




```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python siftdemo.py
SIFT algorithm
key points in image1 1891
key points in image2 1734
matches number 553
detection time cost 0.5321 秒
matching time cost0.0163 秒
```

SIFT 成功 match 了 553 个点。

（2）SuperPoint 结果：



```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python spdemo.py
spdemo.py:9: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default
es the default pickle module implicitly. It is possible to construct malicious pickle data which will
y code during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-model
ls). In a future release, the default value for `weights_only` will be flipped to `True`. This limits
at could be executed during unpickling. Arbitrary objects will no longer be allowed to be loaded via t
they are explicitly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend y
 `weights_only=True` for any use case where you don't have full control of the loaded file. Please ope
tHub for any issues related to this experimental feature.
  model.load_state_dict(torch.load(checkpoint_path))
图像1特征点数量: 1948
图像2特征点数量: 2056
成功匹配的特征点数量: 436
```

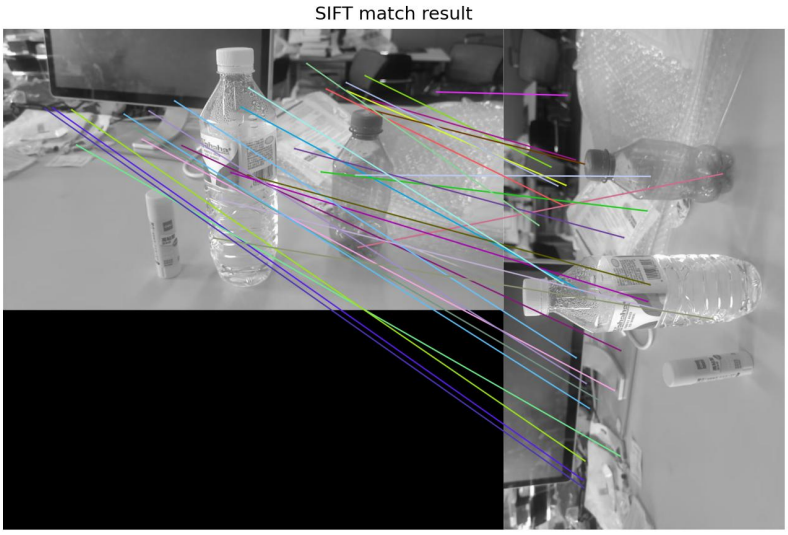
SuperPoint 成功匹配了 436 个特征点。

2、对于同一场景，不同旋转
原图如下：





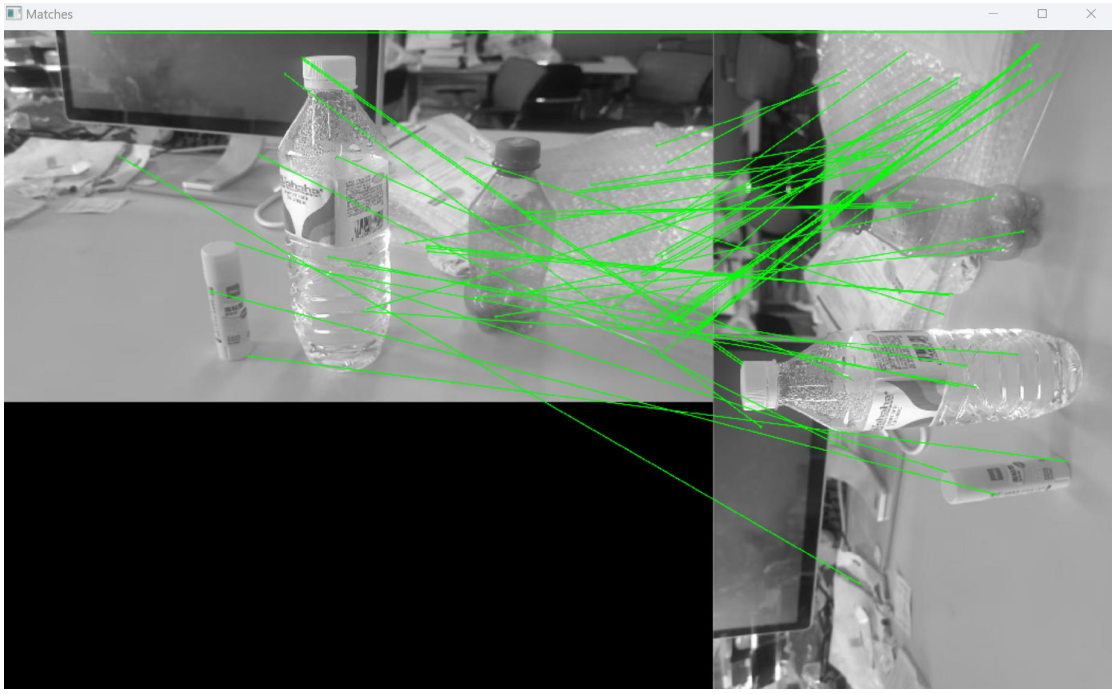
（1）SIFT 结果




```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python siftdemo.py
SIFT algorithm
key points in image1 1845
key points in image2 1844
matches number 1507
detection time cost 0.4479 秒
matching time cost0.0247 秒
```

SIFT 成功匹配了 1507 个点。

(2) SuperPoint 结果



```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python spdemo.py
spdemo.py:9: FutureWarning: You are using `torch.load` with `weights_only=False` (the current default value
es the default pickle module implicitly. It is possible to construct malicious pickle data which will execu
y code during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECURITY.md#untrusted-models for
ls). In a future release, the default value for `weights_only` will be flipped to `True`. This limits the f
at could be executed during unpickling. Arbitrary objects will no longer be allowed to be loaded via this m
hey are explicitly allowlisted by the user via `torch.serialization.add_safe_globals`. We recommend you st
`weights_only=True` for any use case where you don't have full control of the loaded file. Please open an
tHub for any issues related to this experimental feature.
  model.load_state_dict(torch.load(checkpoint_path))
图像1特征点数量: 2001
图像2特征点数量: 1913
成功匹配的特征点数量: 1078
```

SuperPoint 成功匹配了 1078 个点

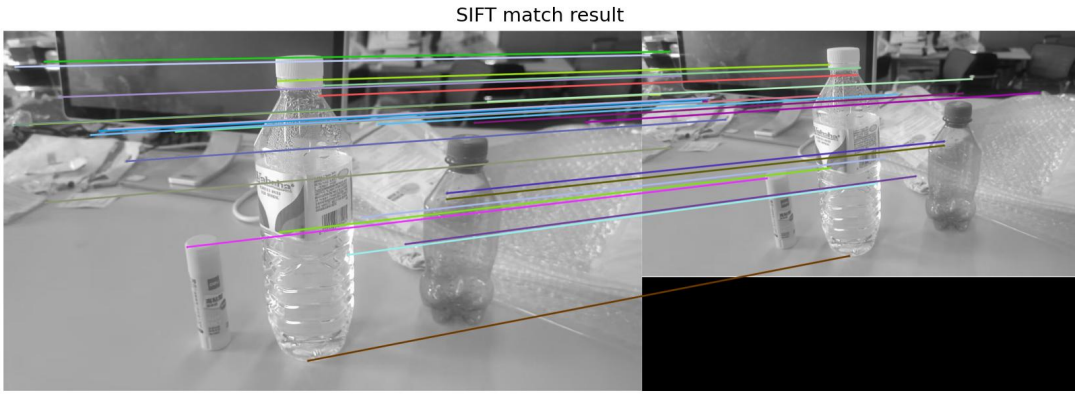
3、对于同一场景，不同缩放

原图如下：





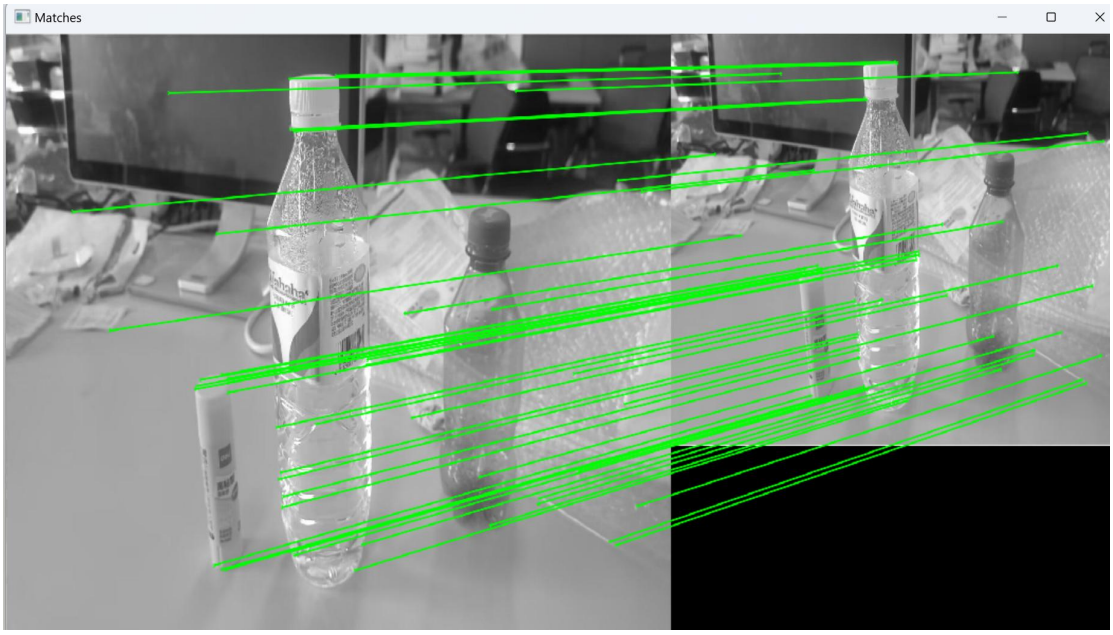
(1) SIFT 结果




```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python siftdemo.py
SIFT algorithm
key points in image1 1845
key points in image2 1265
matches number 970
detection time cost 0.3077 秒
matching time cost0.0210 秒
```

SIFT 成功匹配 970 个特征点

(2) SuperPoint 结果



```
(computer_vision) D:\大学\大三下\计算机视觉\实验\ex7\SuperPoint-master>python s
spdemo.py:9: FutureWarning: You are using `torch.load` with `weights_only=False
es the default pickle module implicitly. It is possible to construct malicious
y code during unpickling (See https://github.com/pytorch/pytorch/blob/main/SECU
ls). In a future release, the default value for `weights_only` will be flipped
at could be executed during unpickling. Arbitrary objects will no longer be all
they are explicitly allowlisted by the user via `torch.serialization.add_safe_g
 `weights_only=True` for any use case where you don't have full control of the
tHub for any issues related to this experimental feature.
  model.load_state_dict(torch.load(checkpoint_path))
图像1特征点数量: 2001
图像2特征点数量: 1456
成功匹配的特征点数量: 1088
```

SuperPoint 成功匹配 1088 个检测点。

总结：综合上述实验结果，SIFT 与 SuperPoint 在同样的输入下，对比成功匹配特征点数量。对于同一场景，视角变化，旋转的情况下 SIFT 更优；而缩放情况下 SuperPoint 更优。